UNIT42 · FIRMWAREX

Firmware Security Assessment

Sample firmware

Generated 11/7/2026, 7:16:20 PM

Executive summary

This assessment recorded **17 findings** across 2 audited binaries and 6 filesystem entries. Analysis status: **complete**.

Results describe the analysis performed on this image. They do not establish that the firmware is secure or that every candidate is exploitable. Severity and confidence are separate; no CVSS score is inferred.

Critical	High	Medium	Low	Info
0	1	6	10	0

Scope and coverage

Stage	Outcome	Restriction
extract	extracted	See scope limits below
static	completed	See scope limits below
emulation	completed	See scope limits below
symbolic	completed	See scope limits below
secrets crypto	completed	See scope limits below
normalize	completed	See scope limits below

Untrusted execution: blocked_pending_mentor_approval. Raw artifact access: disabled.

- Only registered controlled images execute; other images receive static and scanner analysis.
- Unknown path components are pseudonymized to prevent proprietary-string disclosure.
- Symbolic findings are bounded candidates for human review.

- Only executed stages contribute evidence. Blocked, skipped or incomplete analysis is not a clean bill of health.
- This is an automated firmware-image assessment, not a live-device or network scan.
- Recommendations are general review guidance, not verified fixes.

Binary protection summary

Binary	Architecture	NX	PIE	RELRO	Canary	FORTIFY
/bin/service-probe	x86_64	Present	Not detected	partial	Present	Not detected
/bin/taint-demo	x86_64	Present	Not detected	partial	Not detected	Not detected

Detailed findings

---

1. Memory safety candidate High

Property	Result
CWE	CWE-787
Affected file	/bin/taint-demo
Architecture	x86_64
Confidence	probable
Review status	new
Address	0x500008
Function / sink	strcpy
Detected	2026-11-07T13:45:04.661352Z

Evidence

Observation	Value
condition satisfiable	true

Observation	Value
Observed taint path	0x401136 → 0x401040 → 0x500010 → 0x40115c → 0x40116b → 0x401030 → 0x500008
Reaching condition (diagnostic expression)	[{"op":"UGE","args":[{"op":"Extract","args":[247,240, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract","args":[247,240, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":126,"bits":8}]},{"op":"UGE","args": [{"op":"Extract","args":[239,232, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract" … [excerpt; full expression in JSON export]
fault address	0x404030
overflow offset	8
stack hash	bfc322b46d82910dab6757cff82dd695801b2df2172325cb b73b4e88d0926fc2

Evidence reference: evidence:a9fcb179e2fd0726c7d0c66dd46b1acb1672d27ac9aaf00120d704296433ec2b. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

---

2. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established

Property	Result
Function / sink	Not applicable
Detected	2026-11-07T13:45:05.078082Z

Evidence

Observation	Value
rule	credential
fingerprint	9bbc1002715818933b59a9546be453b83c750021991c0b5d4711ffbd0ca9d87d
byte offset	9
byte length	19

Evidence reference: evidence:16cac48bf8027e4bc6cf283910f6b54e36b107a1e1cbfcfb7cbd1ae6b21043bd. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

3. Embedded secret candidate Medium

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-11-07T13:45:05.078286Z

Evidence

Observation	Value
rule	credential
fingerprint	9f19e2f2612304e746a6dce748975c33d8a828ca6cfe3bf61a13f6450b26a167
byte offset	37
byte length	24

Evidence reference: evidence:96606796e1b4837d60fb8503887e1fc29694e09ec7c8bc89377bd91d97b614ad. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

4. Embedded secret candidate Medium

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-11-07T13:45:05.078425Z

Evidence

Observation	Value
rule	private_key

Observation	Value
fingerprint	ea2591adea7adca09a6e1090057e66c0e20d9f3d59af2e19f801dc30d577a5cf
byte offset	62
byte length	78

Evidence reference: evidence:1d09563113067ef7f8fb626dfefd27f51978b0fe2c396909664e064f72b4a4de. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

5. Weak cryptography candidate Medium

Property	Result
CWE	CWE-328
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-11-07T13:45:05.079411Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	6c3e22a3fb7c6b82cc30e75b1b32a276b1464d063e31aeab1c24c5106b32e065

Observation	Value
byte offset	237
byte length	3

Evidence reference: evidence:8b07a6977da70b11da51f27701f375d806f72112ac766afd96bab4f81250d436. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

---

6. Weak cryptography candidate Medium

Property	Result
CWE	CWE-327
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-11-07T13:45:05.079597Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	8a9021c01d3c432123d8dfa041e83477449c77703e7cf3ecb7d189eee7e50617
byte offset	255
byte length	3

Evidence reference: evidence:dfd1dbf67d70ace48a2ba62406b84f74fa55be5f647d8ef4cd2c5ff1d82119e6. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

---

7. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/redacted-eef9f8d4ffc0e9be
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-11-07T13:45:05.080704Z

Evidence

Observation	Value
rule	credential
fingerprint	a6f9515eeeae85a7aab3632e6857757083cbc034e7e3f44f59ecb1082517d8f1
byte offset	38
byte length	25

Evidence reference: evidence:c39c09e1629c42972d097f86be78f20c4ebc5e6ef0de121452747b1d5cd11680. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

8. Memory safety candidate Low

Property	Result
CWE	CWE-676
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	strcpy
Detected	2026-11-07T13:44:56.854437Z

Evidence

Observation	Value
rule	potentially_dangerous_function_symbol
invocation proven	false
overflow proven	false

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

9. Memory safety candidate Low

Property	Result
CWE	CWE-676
Affected file	/bin/taint-demo
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	strcpy
Detected	2026-11-07T13:44:56.855168Z

Evidence

Observation	Value
rule	potentially_dangerous_function_symbol
invocation proven	false
overflow proven	false

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

10. Embedded secret candidate Low

Property	Result
CWE	CWE-798

Property	Result
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-11-07T13:45:05.079191Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	0a2cd690cf9532a79fe38113a296220a80d32ae8c39fd79d90a3fa10bf0b87b0
byte offset	29
byte length	32

Evidence reference: evidence:40d46a94d38e65a680ee565648f5d9ada1c725da40bfc82e7de93c401626688a. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

11. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64

Property	Result
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-11-07T13:45:05.072000Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	e82fd92bbe827411100936861d483404dd3941f02b46235e3343b8d848dcc169
byte offset	511650
byte length	50

Evidence reference: evidence:fda68496d36d3eee1bdf67b2d9b3c7820459fe3bf1a8ecee799385978eda2db7. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

12. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new

Property	Result
Address	Not established
Function / sink	Not applicable
Detected	2026-11-07T13:45:05.072287Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	57f08f1dc2c9ef44f8afc4de304c5bf3fb3e6ded414c335258f60ca82c78b030
byte offset	591137
byte length	26

Evidence reference: evidence:461f7429a03c6a9cd85bdf74584e108c6b9f6e49e294b82caf3721ca1162aa01. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

13. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable

Property	Result
Detected	2026-11-07T13:45:05.072620Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	39bc8bb778d9812068108ac02fb590c1c48a88522252d36d0540925d69d6cf79
byte offset	591169
byte length	26

Evidence reference: evidence:044d3507bba77397f829e2d1d01b83daa265333053e9ad4bbd3db016ae12013a. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

14. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-11-07T13:45:05.072773Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	201d0e99e76582dcb7459b886e96d8e2ebc5dc86bd5fb79f87398b6ba30b7a88
byte offset	591328
byte length	36

Evidence reference: evidence:966c130c5caf2e7ae7b76b414588e82354043affee694e8655d8882aba8a15cb. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

15. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-11-07T13:45:05.072960Z

Evidence

Observation	Value
rule	high_entropy

Observation	Value
fingerprint	1059d1cbd3861010b8b3979ca1cf99c3b89ebdca5ab3f264fc3b2e576cd94c17
byte offset	591392
byte length	36

Evidence reference: evidence:3ced993cb98b227e94255c6581ca89be3548f459cdd5efb603420ebb8ac98ab3. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

16. Embedded secret candidate Low

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-11-07T13:45:05.078978Z

Evidence

Observation	Value
rule	certificate
fingerprint	dac28623b0c9c24a69cf859fd5df093a7f1182ba1c597d1790d755b2102d3934

Observation	Value
byte offset	141
byte length	75

Evidence reference: evidence:447e657d91706aaeefaaf1b8d6a3d38ea221cb72327c0837c6fb13519a55d921. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

---

17. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-11-07T13:45:05.079324Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	aca6f431fff5d2ee54788d94384f5a410b64e9b92e87f0c8f0322c1e43a58feb
byte offset	259
byte length	29

Evidence reference: evidence:078a46522cb2f0586f7664011c8a370e1123e4280d1299111cba5eb34d30bcb1. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

Assessment identity

Property	Value
Firmware ID	69238bfa-5104-4345-b72e-0f634e2193ca
SHA-256	a71d1e4937d2eb468ea965462c27f6b3c99eb2207fa39bd59d04161bf45856c9
Job ID	e07de6b8-95a3-45b8-8d74-7f222d04c6a9
Image bytes	339968